

Problem

A coil with impedance $8 + j6 \, \Omega$ is connected in series with a capacitive reactance X . The series combination is connected in parallel with a resistor R . Given that the equivalent impedance of the resulting circuit is $5 \angle 0^\circ \, \Omega$, find the value of R and X .

Step-by-step solution

Step 1 of 4

A circuit with the impedance, capacitive reactance and resistor is shown in Figure 1.

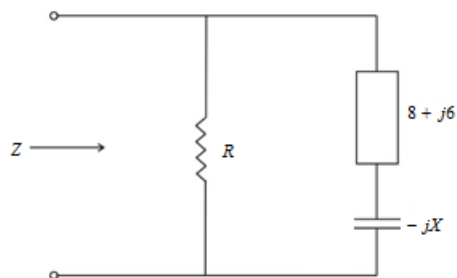


Figure 1

Step 2 of 4

From Figure 1, the equivalent impedance of the circuit is,

$$\begin{aligned} Z &= R \parallel [(8 + j6) - jX] \\ &= R \parallel [8 + j(6 - X)] \\ &= \frac{R[8 + j(6 - X)]}{R + 8 + j(6 - X)} \end{aligned}$$

Step 3 of 4

Substitute $5 \angle 0^\circ$ for Z in the equation.

$$\begin{aligned} 5 \angle 0^\circ &= \frac{R[8 + j(6 - X)]}{R + 8 + j(6 - X)} \\ 5R + 40 + j(30 - 5X) &= 8R + j(6R - XR) \end{aligned}$$

Equate real parts in the equation.

$$\begin{aligned} 5R + 40 &= 8R \\ 3R &= 40 \\ R &= \frac{40}{3} \\ R &= 13.33 \, \Omega \end{aligned}$$

Step 4 of 4

Equate the imaginary parts.

$$(30 - 5X) = (6R - XR)$$

Substitute 13.33Ω for R .

$$30 - 5X = 6(13.33) - 13.33X$$

$$8.33X = 50$$

$$X = \frac{50}{8.33}$$

$$X = 6 \Omega$$

Hence, the values of R and X are 13.33Ω and 6Ω respectively.